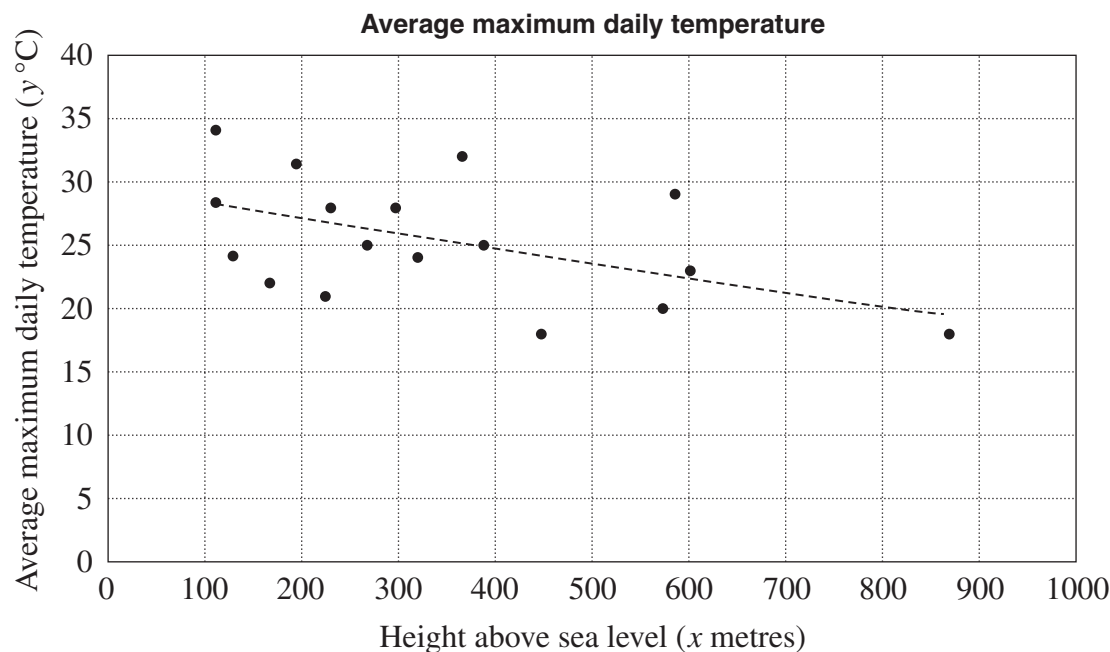


Question 33 (4 marks)

For a sample of 17 inland towns in Australia, the height above sea level, x (metres), and the average maximum daily temperature, y ($^{\circ}\text{C}$), were recorded.

The graph shows the data as well as a regression line.



The equation of the regression line is $y = 29.2 - 0.011x$.

The correlation coefficient is $r = -0.494$.

- (a) (i) By using the equation of the regression line, predict the average maximum daily temperature, in degrees Celsius, for a town that is 540 m above sea level. Give your answer correct to one decimal place.

1

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- (ii) The gradient of the regression line is -0.011 . Interpret the value of this gradient in the given context.

2

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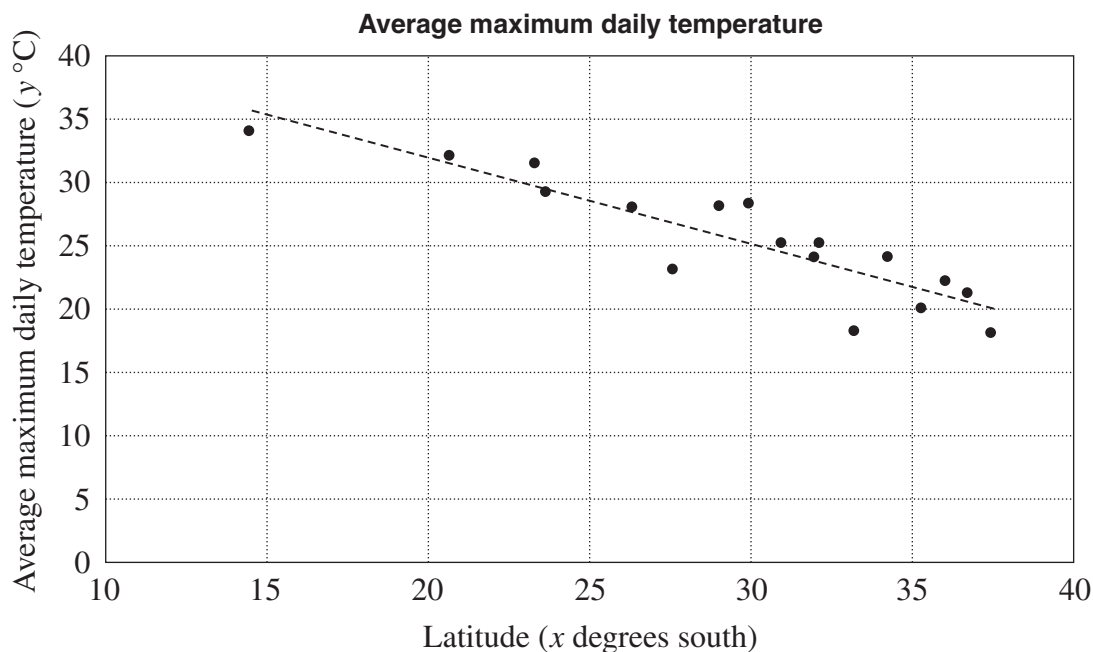
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Question 33 continues on page 29

Question 33 (continued)

- (b) The graph below shows the relationship between the latitude, x (degrees south), and the average maximum daily temperature, y ($^{\circ}\text{C}$), for the same 17 towns, as well as a regression line.

1



The equation of the regression line is $y = 45.6 - 0.683x$.

The correlation coefficient is $r = -0.897$.

Another inland town in Australia is 540 m above sea level. Its latitude is 28 degrees south.

Which measurement, height above sea level or latitude, would be better to use to predict this town's average maximum daily temperature? Give a reason for your answer.

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End of Question 33

Proceed to Booklet 2 for Questions 34–41